University of Ottawa

Department of Mathematics and Statistics

MAT 1302C : Mathematical Methods II

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First Midterm Exam – Version A

October 11, 2012

Surname _____ First Name _____

Student # _____ DGD (Tuesday/Thursday) _____

Instructions:

(a) You have 80 minutes to complete this exam.
(b) The number of points available for each question is indicated in square brackets.
(c) All work to be considered for grading should be written in the space provided. The reverse side of pages is for scrap work. If you find that you need extra space in order to answer a particular question, you should continue on the reverse side of the page and indicate this **clearly**. Otherwise, the work written on the reverse side of pages will not be considered for marks.
(d) Write your student number at the top of each page in the space provided.
(e) No notes, books, scrap paper, calculators or other electronic devices are allowed.
(f) You are strongly recommended to write in **pen**, not pencil.
(g) You may use the last page of the exam as scrap paper.

Good luck!

Please do not write in the table below.

Question	1	2	3	4	5	6	Total
Maximum	4	4	3	3	5	5	24
Grade							

1. [**4 points**] Determine if the following linear system is consistent or inconsistent. (If the system is consistent, you do not need to find the general solution.)

$$\begin{cases} x_1 + 2x_2 = x_3 - 5x_4 + 1 \\ \quad -2x_2 = 4x_4 + 1 \\ 2x_1 + 2x_2 = 2x_3 - 6x_4 + 4 \end{cases}$$

Solution:

We write the augmented matrix of the system and row reduce it

$$\left[\begin{array}{cccc|c} 1 & 2 & -1 & 5 & 1 \\ 0 & -2 & 0 & -4 & 1 \\ 2 & 2 & -2 & 6 & 4 \end{array}\right] \xrightarrow{R_3 \to R_3 - 2R_1} \left[\begin{array}{cccc|c} 1 & 2 & -1 & 5 & 1 \\ 0 & -2 & 0 & -4 & 1 \\ 0 & -2 & 0 & -4 & 2 \end{array}\right] \xrightarrow{R_3 \to R_3 - R_2} \left[\begin{array}{cccc|c} 1 & 2 & -1 & 5 & 1 \\ 0 & -2 & 0 & -4 & 1 \\ 0 & 0 & 0 & 0 & 1 \end{array}\right]$$

$$\xrightarrow[R_2 \to R_2 - R_3]{R_1 \to R_1 - R_3} \left[\begin{array}{cccc|c} 1 & 2 & -1 & 5 & 0 \\ 0 & -2 & 0 & -4 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{array}\right] \xrightarrow{R_2 \to -\frac{1}{2}R_2} \left[\begin{array}{cccc|c} 1 & 2 & -1 & 5 & 0 \\ 0 & 1 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{array}\right]$$

$$\xrightarrow{R_1 \to R_1 - 2R_2} \left[\begin{array}{cccc|c} 1 & 0 & -1 & 1 & 0 \\ 0 & 1 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{array}\right]$$

Since the rightmost column is a pivot column, the linear system is inconsistent.

2. [**4 points**] Determine all values of h such that the vector $\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$ belongs to Span $\left\{ \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 3 \\ 1 \\ -2 \end{bmatrix}, \begin{bmatrix} -1 \\ h \\ 1 \end{bmatrix} \right\}$.

Solution: We write the corresponding augmented matrix and then row reduce it:

$$\left[\begin{array}{ccc|c} 1 & 3 & -1 & 0 \\ -1 & 1 & h & 1 \\ 0 & -2 & 1 & 0 \end{array}\right] \xrightarrow{R_2 \to R_2 + R_1} \left[\begin{array}{ccc|c} 1 & 3 & -1 & 0 \\ 0 & 4 & h-1 & 1 \\ 0 & -2 & 1 & 0 \end{array}\right] \xrightarrow{R_3 \to R_3 + \frac{1}{2}R_2} \left[\begin{array}{ccc|c} 1 & 3 & -1 & 0 \\ 0 & 4 & h-1 & 1 \\ 0 & 0 & \frac{1}{2}h + \frac{1}{2} & \frac{1}{2} \end{array}\right]$$

The vector the vector $\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$ belongs to Span $\left\{ \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 3 \\ 1 \\ -2 \end{bmatrix}, \begin{bmatrix} -1 \\ h \\ 1 \end{bmatrix} \right\}$ if and only if is the latter matrix is consistent, that is, the rightmost column is not a pivot column. This is equivalent to $\frac{1}{2}h + \frac{1}{2} \neq 0$, or $h \neq -1$.

3. **[3 points]** Find the general solution for the matrix equation $A\mathbf{x} = \mathbf{b}$, where

$$A = \begin{bmatrix} 1 & -2 & 0 & 1 \\ -2 & 4 & 1 & 1 \end{bmatrix} \quad \text{and} \quad \mathbf{b} = \begin{bmatrix} 1 \\ -4 \end{bmatrix}.$$

You should write your solution in **vector parametric form**.

Solution: Assume $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$. We row reduce the augmented matrix of the corresponding linear system.

$$\left[\begin{array}{cccc|c} 1 & -2 & 0 & 1 & 1 \\ -2 & 4 & 1 & 1 & -4 \end{array}\right] \xrightarrow{R_2 \to R_2 + 2R_1} \left[\begin{array}{cccc|c} 1 & -2 & 0 & 1 & 1 \\ 0 & 0 & 1 & 3 & -2 \end{array}\right]$$

x_1, x_3 basic. x_2, x_4 free. Therefore the general solution is:

$$\begin{cases} x_1 = 2x_2 - x_4 + 1 \\ x_2 = \text{free} \\ x_3 = -3x_4 - 2 \\ x_4 = \text{free} \end{cases}$$

and the vector parametric form of the solution is

$$\mathbf{x} = \begin{bmatrix} 2x_2 - x_4 + 1 \\ x_2 \\ -3x_4 - 2 \\ x_4 \end{bmatrix} = x_2 \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} -1 \\ 0 \\ -3 \\ 1 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \\ -2 \\ 0 \end{bmatrix}$$

4. **[3 points]** For which values of the parameters a and b the linear system given below is inconsistent?

$$\begin{cases} x_1 + 2x_2 = 1 \\ x_2 - 2x_3 = 0 \\ -2x_1 + ax_2 - x_3 = b \end{cases}$$

Solution: We row reduce the corresponding augmented matrix:

$$\left[\begin{array}{ccc|c} 1 & 2 & 0 & 1 \\ 0 & 1 & -2 & 0 \\ -2 & a & -1 & b \end{array}\right] \xrightarrow{R_3 \to R_3 + 2R_1} \left[\begin{array}{ccc|c} 1 & 2 & 0 & 1 \\ 0 & 1 & -2 & 0 \\ 0 & a+4 & -1 & b+2 \end{array}\right] \xrightarrow{R_3 \to R_3 - (a+4)R_2} \left[\begin{array}{ccc|c} 1 & 2 & 0 & 1 \\ 0 & 1 & -2 & 0 \\ 0 & 0 & 2a+7 & b+2 \end{array}\right]$$

The linear system is inconsistent exactly when the rightmost column is a pivot. The only way this can happen is that $a = -\frac{7}{2}$ and $b \neq -2$.

5. [**5 points**] Determine if the following vectors are linearly dependent or linearly independent:

$$\begin{bmatrix}1\\0\\1\end{bmatrix}, \begin{bmatrix}-1\\1\\1\end{bmatrix}, \begin{bmatrix}8\\-3\\2\end{bmatrix}$$

If the vectors are linearly dependent, then you should find a (nontrivial) linear combination of them which is equal to the zero vector.

Solution: We should row reduce the following matrix:

$$\begin{bmatrix}1&-1&8\\0&1&-3\\1&1&2\end{bmatrix} \xrightarrow{R_3\to R_3-R_1} \begin{bmatrix}1&-1&8\\0&1&-3\\0&2&-6\end{bmatrix} \xrightarrow{R_3\to R_3-2R_2} \begin{bmatrix}1&-1&8\\0&1&-3\\0&0&0\end{bmatrix}$$
$$\xrightarrow{R_1\to R_1+R_2} \begin{bmatrix}1&0&5\\0&1&-3\\0&0&0\end{bmatrix}$$

Since the third column is not a pivot column, these vectors are linearly dependent.

To find a linear combination, we should solve the following vector equation:

$$x_1\begin{bmatrix}1\\0\\1\end{bmatrix} + x_2\begin{bmatrix}-1\\1\\1\end{bmatrix} + x_3\begin{bmatrix}8\\-3\\2\end{bmatrix} = \begin{bmatrix}0\\0\\0\end{bmatrix}$$

We note the following row reduction (based on the above calculation):

$$\left[\begin{array}{ccc|c}1&-1&8&0\\0&1&-3&0\\1&1&2&0\end{array}\right] \xrightarrow{\text{Row reduction}} \left[\begin{array}{ccc|c}1&0&5&0\\0&1&-3&0\\0&0&0&0\end{array}\right]$$

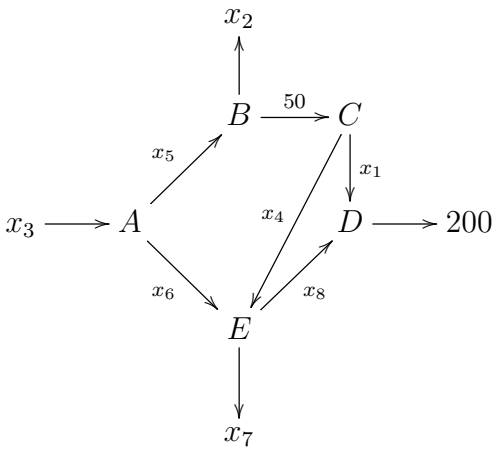
x_1 and x_2 are basic whereas x_3 is free. Therefore the general solution is:

$$\begin{cases}x_1 = -5x_3\\ x_2 = 3x_3\end{cases}$$

and an example of a linear combination is

$$x_3 = 1 \Rightarrow x_1 = -5, x_2 = 3 \Rightarrow \quad (-5)\begin{bmatrix}1\\0\\1\end{bmatrix} + 3\begin{bmatrix}-1\\1\\1\end{bmatrix} + 1\begin{bmatrix}8\\-3\\2\end{bmatrix} = \begin{bmatrix}0\\0\\0\end{bmatrix}$$

6. Consider the traffic flow described by the following diagram. The letters A through E label intersections. The arrows indicate the direction of flow (all roads are one-way) and their labels indicate flow in cars per minute.



(a) **[3 points]** Write down a linear system describing the traffic flow, i.e., all constraints on the variables $x_i, i = 1, \ldots, 8$. (Do not solve the linear system at this stage.)

Solution:

A	x_3	$= x_5 + x_6$
B	x_5	$= x_2 + 50$
C	50	$= x_1 + x_4$
D	$x_1 + x_8$	$= 200$
E	$x_4 + x_6$	$= x_7 + x_8$
Total	x_3	$= x_2 + x_7 + 200$

(b) [**1 point**] The reduced row echelon form of the augmented matrix corresponding to the linear system in part (a) is:

$$\left[\begin{array}{cccccccc|c} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 200 \\ 0 & 1 & 0 & 0 & -1 & 0 & 0 & 0 & -50 \\ 0 & 0 & 1 & 0 & -1 & 0 & -1 & 0 & 150 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & -1 & -150 \\ 0 & 0 & 0 & 0 & 0 & 1 & -1 & 0 & 150 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

Find the general flow pattern (i.e., the general solution of the linear system).

Solution: x_1, x_2, x_3, x_4, x_6: basic; x_5, x_7, x_8: free.

$$\begin{cases} x_1 = -x_8 + 200 \\ x_2 = x_5 - 50 \\ x_3 = x_5 + x_7 + 150 \\ x_4 = x_8 - 150 \\ x_5 = \text{free} \\ x_6 = x_7 + 150 \\ x_7 = \text{free} \\ x_8 = \text{free} \end{cases}$$

(c) [**1 point**] Suppose that due to road work, the flow along ED is limited to a maximum of 200 cars per minute. What is the maximum possible flow along CE?

Solution: Since $x_4 = x_8 - 150$, from $x_8 \leq 200$ it follows that $x_4 \leq 50$, i.e., the maximum number of cars along CE is 50 cars per minutes.